Presentation & Evaluation of LOCARN: Low Opex and Capex Architecture for Resilient Networks

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Abstract. *This article proposes LOCARN: an alternative network architecture providing a packet connectivity layer, which is able to self-adapt its routing paths opportunistically according to both the network resources changes and the effective traffics fluctuations. Moving close to a global maximization of available resources usage and assuming high resiliency under failures, LOCARN focuses on components coupling simplicity and plug-and-play guidance. The following article presents the architecture and then evaluates its performances. Specifically, LOCARN produces significant overheads in counterparty of its suitable properties, because of the radicality of its design. Considering its overall characteristics, we envision LOCARN as an alternative to current packet transport network architectures based on IP/MPLS, typically metropolitan and cores subparts of operators' networks. We show thereafter that in a such use case the overhead impact is acceptable and even negligible beside the transmission capacities available in such networks. To do so, we rely on analysis, on computer simulations and finally on a statistical model.*

Keywords: *packet transport network, flat network architecture, self-* properties, scalability evaluation*

1. Introduction

Over the last decade, complexity has become a more and more decisive issue for network operators, as well as the Operational EXPenditure (OPEX) costs that are more or less directly involved by this latter. Along the evolution of operators networks technologies – among which the most decisive is probably the transition to packets transmissions [1] – the control of networks have progressively evolved from the legacy purely centralized management to more hyrid control designs combining distributed control planes which collaborate with centralized controllers. The rise of distributed control planes (historically: ATM, IP and finally IP/MPLS) constitutes a major step of the operators' networks automation, that should go toward a management simplification. Yet operators still spend a lot of time and resources to continually adapt their networks to the traffics demands through three main steps: sizing, planification and configuration of protection mechanisms. To go further toward simpler management, several proposals emerged during the recent years. We notice three main trends:

– *Path Computation Element (PCE)* architectures that propose to add upon IP/MPLS some devices dedicated to take some complexe decision based on an overall view of the network, in collaboration with an Interne Gateway Protocol extended for Traffic Engineering (e.g. ISIS-TE [LS08]) to get topological information and a signaling protocol (e.g. RSVP-TE [ABG+01]) to apply decisions.

– *Autonomic Networking* field that involved many academic contributions under the form of frameworks and architectures (well surveyed in [MALP12]). Yet from those proposal result almost no concrete deployments among operators' networks (except in the mobile access with 3GPP normalization, under the terms of "Self-Organizing Networks").

– *Software Define Networking (SDN)* field that mainly aims to increase the network flexibility by allowing advisors to define rules to program the network behavior. In terms of design the SDN approach consists in centralizing the decision making. This is the current big trend of network automation (see notably [NMN+14]).

1. through the generalization of the Internet Protocol

In 2010, a very innovative packet network architecture named LOCARN (i.e. Low Opex and Capex Architecture for Resilient Networks) has been proposed by Orange Labs in the context of the European TIGER2 project [tig] whose one of its objectives aimed at making packet based transport network solution evolving towards higher manageability. To that end, the project has explored several proposals to improve the networks' control and management by focusing on the autonomic networking aspects that would allow to master complexity [GDCR+10a, GDCR+10b]. LOCARN is born from the observation that networks automation trend-setting approaches mentioned above involve a significant collection of components and mechanisms in nodes: protocols, tables, path computation module, label/bandwidth allocation, mechanisms for protection. Based on this observation, LOCARN proposes to radically revisit the way to think network automation by a new design of its (own) data and routing plane in order to guarantee a high level of *simplicity*, *self-adaptation* and *resiliency*. The unpleasant aspect those three properties finally lies on the amount of non client packets produced by the architecture, that may possibly present some performances and scalability issues.

First, we present the LOCARN mechanisms, to make understand its potential and to notice its technical challenges (section II). Then, in the rest of the document, we study LOCARN scalability and performances in normal conditions in section III, and in condition of failures in section IV, focusing in particular on estimations of the architecure's overheads costs. Finally, the section V concludes about the results of this article.

2. The LOCARN Architecture

2.1. *Technical Overview*

LOCARN is a flat packet network architecture providing point-to-point bidirectional communications through the definition of its own data plane and routing plane mechanisms, in a complete agnostic way with regard to the IP stack. The functional architecture (illustrated in Fig.1) is composed of two kinds of functional nodes: (i) *Edges Nodes* (ENs) that constitute ingress/egress nodes of a LOCARN transport domain (ii) *Transit Nodes* (TNs) that solely operate packet forwarding.

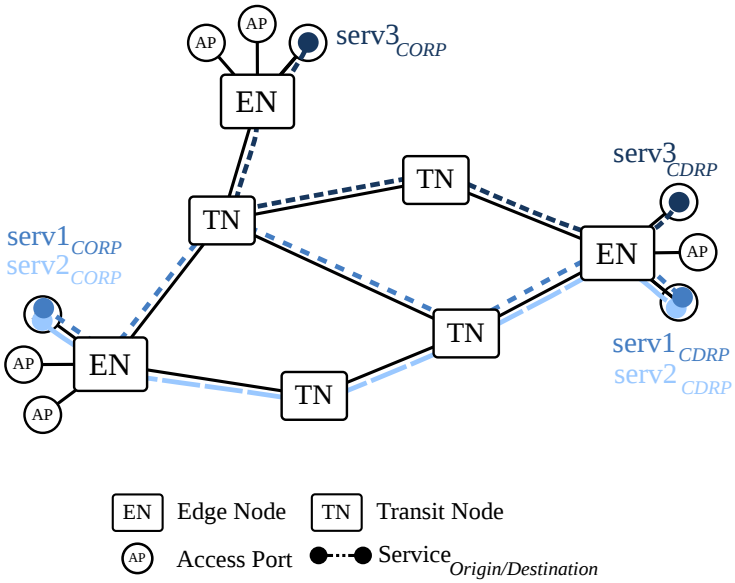


Figure 1: The LOCARN functionnal architecture on a small example

The functional architecture also defines: (i) *Access Ports* (APs) that are external ENs ports (they can be physical or logical ports according to the implementation guidance) (ii) *Services* that are end-to-end bidirectional channels [2]. Services provide customer data with the transport of its client information between two APs across the domain. They are composed by two Reference Points (RPs) sharing a common service identifier: a Connection Origin RP (CORP) and a Connection Destination RP (CDRP). The only management operations required in LOCARN is registration/unregistration of RPs using unequivocal services names among the domain. Then LOCARN is in charge of the establishment, maintenance and adaptation of declared services over time.

The LOCARN solution concept relies on three network mechanisms:

2. the LOCARN architecture forms a layer network in the sense of the ITU-T G.805 recommendation: “A topological component that represents the complete set of access groups of the same type which may be associated for the purpose of transferring information"; in the ITU transport terminology, a LOCARN service is a point-to-point *transport entity*, i.e.: “an architectural component which transfers information between its inputs and outputs within a layer network"

Autoforwarding – Autoforwarding designates a packet relaying mechanism that uses exclusively information contained in the packet itself for the determination of the next hop interface. In LOCARN, the data packets are autoforwarded from end-to-end by carrying in its headers the complete sequence of output ports to cross from the origin to the destination (including the AP as the last port). Autoforwarding leads to a drastic simplification of the data plane technical requirements, both in terms of computation and memorizing. Indeed, the autoforwarding header is just added at data packets encapsulation in the ingress EN whereas successive TNs forward packets by obtaining the output port directly from this header. A sequence cursor is incremented at each step of the course to indicate the header position to read. An autoforwarding data plane does not require to store any Forwarding Information Base (FIB) in TNs neither a fortiori to perform any table lookup operation.

Enhanced flooding – To obtain the autoforwarding header information included at the data packets encapsulation, an EN runs a source routing process for each CORP registration. A LOCARN source routing process can be decomposed into six steps: (i) a service path request is flooded across the network; (ii) over time, some request messages reach the EN where the corresponding CDRP is registered (arriving from distinct end-to-end paths); (iii) for each request message, the "destination EN" returns an answer message to the "origin EN" [3]. Each answer message is carrying a distinct path autoforwarding information to the origin EN (each request message stores the crossed output ports along its course); (iv) along the backward course, answer messages take the opportunity to collect statistics about the crossed interfaces (typically bandwidth usage, queuing usage...); (v) then the origin EN typically receives numerous packets sent by the destination which each of them contains both the autoforwarding information and statistics about the path. Let us notice that loops are intrinsically prevented during the flooding propagation by deleting packets that followed a cycle ; (vi) finally the EN origin selects the best path by successive comparisons between the arriving path and the one that is currently used. In LOCARN, source routing is thus purely achieved through paths *discovery* and *se-*

3. answers are autoforwarded along the corouted reverse paths from destination to origin, each request message stores the crossed input ports along its course

lection. No routing algorithm is involved, no routing computation or routing tables either, and no convergence times: a path has changed from the moment that the encapsulation function has been modified.

The decisive specificity of LOCARN routing lies on the fact that: (i) a routing process is performed for each service – “serv1" and “serv2" illustrated this in Fig.1 ; (ii) among the (large) amount of discovered paths, a service will typically favors the less used paths. To do so, the service's path selection is partially or totally based on the discovered paths' meta-information collected (i.e. interfaces statistics about bandwidth consumption and queuing loads), which actually reflects the level of effective resources usage along the path [4]; (iii) for each service, the source routing steps are relaunched periodically to adapt the active path over time to any kind of noticeable evolution between the endpoints (referred as “path-optimizations"). Thus, on one hand each service self-adapts periodically according to its perception of the network state between its endpoints; and on the other hand a service adaptation is itself noticeable by other services within the domain, and may trigger their own adaptation [5]. Finally, over successive path-optimizations, LOCARN tends hollistically toward a global, opportunistic and very adaptive distribution of client traffics among the domain without involving any complexity in the achitectural design. In-depth studies regarding LOCARN dynamic behaviors are out of the scope of this document – both positive behaviors like overall path arrangements and negative aspects like possible erratic behaviors.

End-to-end fault detection – The periodical relaunch of source routing steps exposed above make services able to recover if the connectivity has been lost for any reason. Yet, the optimization periodicity is envisioned at least around several seconds making the recovery time relatively long. With regards to the operators Transport Networks requirements in terms of resiliency, each LOCARN active path is continously checked by very frequent round trips of specific small OAM

4. several routing policies can be implemented. The use of the available bandwidh criteria tends to balance loads for example whereas the end-to-end delay criteria can be used to provide adapted path for clients traffics sensitive to delay (on its side, a path end-to-end delay is obtained by noticing the Round Trip Time (RTT) at the path answer arrival time).

5. a service adaptation leads to a more or less significant redirection of the client traffic which impacts interfaces statistics that will be collected by future services' discoveries

packets (i.e. Operation And Maintenance); the lack of three OAM answers at the CORP means that a fault has occured. These packets exchanges, initiated by CORPs, are strictly transfered like data packets with autoforwarding. The interval of OAM sends determines the detection time, it must be envisionned around several milliseconds to reach carrier grade networks.

2.2. *LOCARN Benefits: Properties & Behaviors*

We assume that in an operator viewpoint, the overall interest of their Transport Networks automation are: (i) the simplification of connectivity provisioning through limiting complex planning and design processes ; (ii) the efficiency of network resource's utilization especially through dynamic shared restoration in replacement of dedicated static protection ; (iii) rapid service turn-up through the timely support of bandwidth on-demand services which allow some new revenue opportunities. In such perspectives, LOCARN exhibits some advantageous properties and behaviors:

LOCARN is simple

LOCARN proposes simple nodes design based on minimal computation and storage: low states in edge nodes and no state at all in transit nodes. Hence the architecture is easy to implement into software or hardware components with low technical requirements. Furthermore LOCARN is simple to operate: the addition of new communications simply consists in the declaration of the appropriate services' Reference Points. Once it is established, each service is included in the network domain adaptation process without requiring any planning-configuring step. The LOCARN operational simplicity is known as “plug-and-play", because the network is ready to work as soon as the devices are plugged onto the underlying layer whereas a service is ready to transmit immediately from the moment its two endpoints have been declared. The simplicity is besides a mean to reduce the financial costs: the nodes low technical prerequisites involve a minimum CAPital EXpenditure (CAPEX) whereas the operational simplicity leads to minimize the OPerationnal EXPenditure (OPEX). Finally, the data plane simplicity con-

stitutes a significant potential for energy saving because both Forwarding Information Bases (FIBs) and look-up operations are completely eliminated for packet switching along the transit.

LOCARN assumes resources resiliency

LOCARN copes with resources failures, ensuring the continuity of client information transit with the minimum of impact. The LOCARN resiliency relies on a purely reactive restoration process without any preliminary configuration of any protection path. On a service fault detection (related to the settings of OAM exchanges), the CORP simply re-launches the source routing processes in order to restore the path. This reactive approach is possible because the path discovery is very quick (a path between an origin and a destination is basically obtained during its Round Trip Time). Doing so, LOCARN avoids the waste of network transmission capacity in regard to the static dedicated protection, moreover it is intrinsically able to cope with multiple failure uses cases, which is a major distinction with protection mechanisms used in MPLS/GMPLS based solutions (see “Fast Reroute" [PSA05]). Since an end-to-end path can be discovered toward the destination, it constitutes a potential restoration path. Hence, in LOCARN the simple resource addition (links, nodes) within a domain increases the network robustness and the services' availability. In terms of operational implications, this completely reactive approach greatly simplifies maintenance by preserving the “plug-and-play" property. For example, a node isolation that is usually a heavy and error prone operation in transport technologies simply consists in deplug it and replug it. The degree of resiliency is tunable according to the frequency of OAM exchanges.

LOCARN assumes clients dynamicity

LOCARN has been conceived to handle dynamic clients transportation demands. Indeed, whether because of an irregular bandwidth demand of services or because of an irregular arrival or departure of these latter, routing paths are adapting among the LOCARN domain according to the effective observed consumption. Many communications can be established and released without involving adaptations if they cause a minimum impact on other services whereas the ponctual need for bandwidth of one service may involve several paths shifts for exam-

ple. Simply, a LOCARN layer constantly adapts all paths to satisfy both evolutions of the transportation demands and evolution of the available resources. The degree of dynamicity is tunable according to the frequency of path optimization processes.

LOCARN is a suitable base for Autonomic Networking

According to the terminological distinction exposed in [MALP12], we prefer the “automatic" to the “autonomic" term to qualify LOCARN in so far that in the actual architecture, no High Level Objectives (HLOs) are externally defined to pilot the path adaptations criteria. However, an Autonomic Network Management System (ANMS) can be introduced quite easily for the coordination or ENs decisions on the basis of the topological information they have collected over time. This way, LOCARN can be the mean to introduce autonomic networking principles in an incremental way.

2.3. *LOCARN issues: Overheads and Scalability*

As exposed in the introduction, the LOCARN's counterparty to its simplicity and dynamicity is the amount of overhead messages generated over time. Indeed, unlike other protocols, the LOCARN overhead is not only related to the network dimensions, but it also depends on the amount of services. We distinguish three kinds of overheads corresponding to the three basic LOCARN mechanisms in so far each one involves the transfer of non client information.

Data plane overhead

Because of autoforwarding, the LOCARN data packets header's size depends on the path's lengths, which presents a priori a scalability issue. However, with the LOCARN specific framing scheme, data headers are actually smaller than common packet transport technologies. Indeed, in LOCARN a data header solely contains output ports from the source to the destination but it includes no node identifier neither for source, destination or intermediate nodes. In practice, for a 20 hops path [6] that

6. that can be considered as a very long path across a single transport domain

needs to store 20 output ports, if the implementation assumes for example that the identifier of transit ports is encoded on 1 byte (allowing 256 transit ports per Transit Node), the header remains smaller than a MAC-in-MAC encapsulation of Carrier Ethernet frames (802.1ah) and it remains equivalent to five MPLS label stack entries [RTF+01]. Hence, if data headers should be consider for a very accurate calculation of the total LOCARN overhead, it remains a not critical issue beside the overhead induced by routing and OAM as we shall see hereafter.

Flooding's overhead

The floodings overhead is definitively the major point studied here. Let us first notice that it can not be envisioned to let the flooding process terminates by itself since the number of generated messages would exceed a reasonable number, even for small networks. In LOCARN the floods propagation is simply limited by introducing a Time To Live (TTL) mechanism to path requests messages. This approach raises the question of the TTL tuning. We easily understand that the TTL must be at least equal to the shortest path(s) length between an origin and a destination to allow the establishment of a connectivity; whereas each TTL increment increases the amount of path discovered whose the beneficial consequence is a higher potential for path optimization and for recovery. For simplification, we consider in following sections a global TTL applied to all services path discoveries. The selected value is comprised between the network diameter and a reasonable upper bound to be estimated.

OAM overhead

The OAM exchanges between service's end-points is another critical factor of overhead cost in LOCARN due their high frequency. Indeed, to meet high resiliency level, OAM messages must be sent with high frequency (i.e. around an interval of 10ms or 15ms for high resiliency requirements under 50ms).

2.4. *Related Works*

Conceptual ancestors

The first LOCARN conceptual ancestor is, in the historical order, the Dynamic Source Routing protocol [JHM07]. Conceived for the Mobile Adhoc NETworks, DSR is a routing protocol that aims to minimize the routing messages overhead by considering the resources scarcity of the wireless environment. On the contrary to proactive table driven IP routing protocols (e.g. OSPF) that continuously diffuses control packets among the network, DSR achieves reactive routing from the source on demand. The list of nodes to follow until the destination are collected at the source. Then, the whole path to the destination is specified in packets specific headers for their routing from end-to-end.

The second related work that is conceptually close to LOCARN is the APLASIA architecture [RRB+13]. APLASIA work is inspired from the LOCARN initiative and is notably based on the same primary concepts, namely autoforwarding and routing based on paths discoveries by means of network floods. Yet, in the APLASIA design, those mechanisms are used in a different way, providing ultimately very different properties and behaviors than LOCARN. Among other differences, the Adaptive Probabilistic Flooding [BBC+12] algorithm is employed to minimize the message generation involved by flooding processes (reducing at the same time the amount of discovered paths).

The principal competitors

Several trends exist today for operators transport networks in terms of distributed control planes. The main packet perspective consists in the deployment of IP/MPLS “automation suite" standards. Namely: the ISIS-TE[LS08] protocol for the collection of topological information, the RSVP-TE or LDP [ABG+01, AMT07] protocols for the end-to-end establishment of communications (with or without bandwidth reservation), the Fast Reroute mechanism (FRR) through the RSVP-TE protocol extension [PSA05] for the quick recovery of communications (local and temporary protection), this latter is typically coupled to Bidirectional Forwarding Detection (BFD) [KW10] to reach a quick detection of failures.

Some research efforts toward Ethernet based technologies also exist and aim to provide large scale automated packet networks, under the terms of “layer 2 routing". Among them we distinguish [VMSK13, Per04] that replace the Ethernet Spanning Tree Protocol (STP) by the IP link state routing, and [RTA00] that combine the two.

3. LOCARN's Overhead and Performances in Normal Conditions

In this section we study the LOCARN's overheads cost in normal conditions, it is to say that we do not consider the floodings involved for paths recoveries and due to the infrastructure failures. In a first step, we provide an analytic formula to estimate the number of messages produced by floodings according to the network topology and LOCARN parameters. Then, we verify this formula by simulation and bring out some performances results through a realistic network example. Finally, we explain a method to estimate the OAM overhead cost, and gather results for a global overview of LOCARN overheads upon several networks' profiles.

3.1. *Analytic Formula of Flooding Messages' Production*

The two parameters determining the message generation of a flooding process bounded by a Time To Live (TTL) are the network density (δ_{net}) and the TTL value. Whether a network having N nodes and L links, the mean density corresponds to $\delta_{net} = L \times 2/N$ (the mean number of interfaces per node). The average message generation can be estimated quite easily: indeed the origin node sends on average δ_{net} messages (across all interfaces), whereas successive nodes retransmit on average $\delta_{net} - 1$ until the TTL is reached (across all interfaces except the incoming one). Thereby, whether M_N representing the number of messages generated by duplications at the propagation step N, we have:

$$M_N = \delta_{net} \times (\delta_{net} - 1)^{N-1} \tag{1}$$

Then we get an estimation of the global amount of messages generated over the network (M_{flood}) by summing terms from M_1 to M_{TTL}:

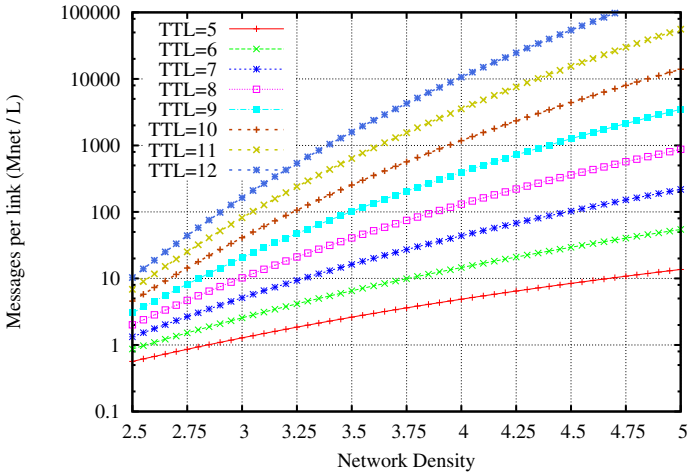


Figure 2: Evaluation of the message generated due to a single flood according to the network density and TTL bound (reduced in messages per link)

$$\begin{aligned} M_{flood}(\delta_{net}, TTL) &\approx \delta_{net} \times \sum_{i=1}^{TTL} (\delta_{net} - 1)^{i-1} \\ &\approx \delta_{net} \times \frac{(\delta_{net} - 1)^{TTL} - 1}{\delta_{net} - 2} \end{aligned} \tag{2}$$

The expression provides an *upper bound* estimation in so far it does not include that, along the flooding propagation, messages which achieved a cycle are discarded. Hence all messages estimated subsequently to loop detections are counted in excess, which yet allows to express the message generation *magnitude* through a simple formula. In Fig. 2 are gathered numerical applications giving an overview of this magnitude reduced per link, by varying TTL (ranging from 5 to 12) and network density δ_{net} (ranging from 2.5 to 5). These results correspond to a numerical application of (2) to a network with $N = 50$ nodes [7], dividing the global amount of messages generated over the network by the number of links ($L = (\delta_{net} \times N)/2$). Unsurprisingly the amount of mes-

7. application for a 100 nodes with the same density gives similar magnitudes, globally low

	Topological Description	**Density (δ_{net})**	**Diameter (D)**
Profile 1	“Scarce Extended"	2.8	10
Profile 2	“Medium"	3.5	7
Profile 3	“Dense Little"	5	5
Profile 4	“Medium Extended"	3.5	10
Profile 5	“Dense Medium"	5	7
Profile 6	“Dense Extended"	5	10

Table 1: Topological profiles parameters for analytic estimations

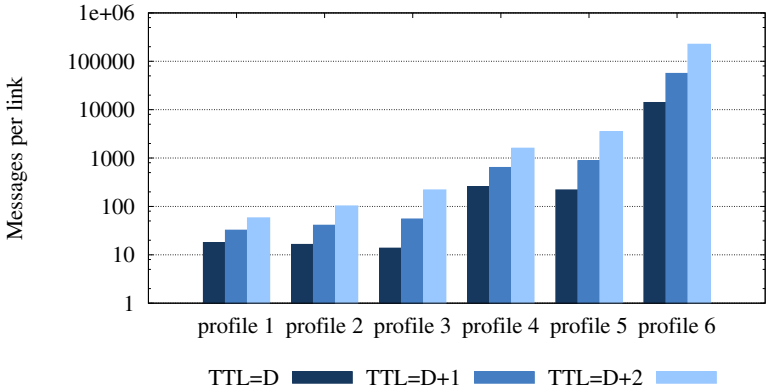


Figure 3: Message generation dut to a single flood for several network profiles and TTL arrangements

sages quickly explodes with the network density, making the overhead cost unacceptable for “dense networks" and “excessive" TTL range. In Fig. 3 are reported the magnitudes of the messages generation for one flood that can be expected per link upon several networks (profiles are exposed in Table 1). The TTL of flooding is chosen to exceed the diameter in order to estimate the most expensive service path discovery. The obtained results bring out that upon “reasonable" network dimensions (profiles 1, 2 and 3) a link transmits on average between ten and hundred of path request messages within a flooding process.

3.2. *Path Discovery: Performances vs Overhead*

In this subsection, we expose some LOCARN’s simulation results. The LOCARN mechanisms (data plane, routing plane and OAM) have

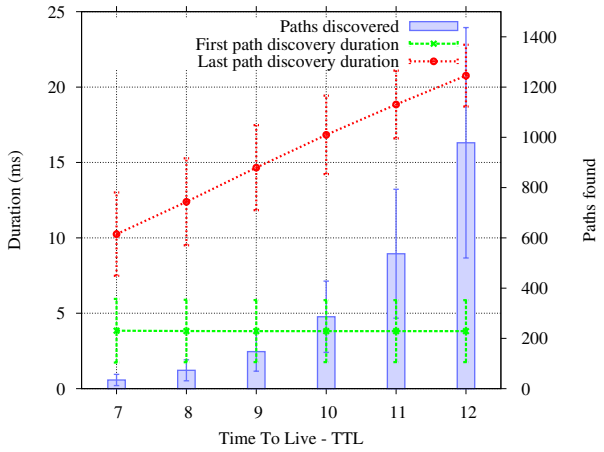


Figure 4: Path discovery performances (average and standard deviation for 1000 randomly picked services)

been implemented in the Omnet++ discrete event simulator. A network infrastructure is simulated by taking into account: the network topology, the links datarates and the propagation delays (packet-processing delays are not implemented here as they are negligible with regard to the propagation ones in the example taken, see [RWW04]).
To study performances and costs over a concrete and realistic use case as those encountered in Orange Group, the results of this section correspond to a LOCARN application to the national (France) core network topology. The latter includes N=42 nodes, L=68 links, has a diameter D=7 hops and a mean density $\delta_{net} = 3.23$ whereas the nodes degree standard deviation is $\sigma_\delta = 1.12$. Propagation delays are implemented in accordance with the optical propagation across the real inter-nodes distances (the mean propagation delay is $445.9\mu s$). Nodes interfaces are associated with unlimited FIFO queues without priorization between data, control or OAM packets.
Hereafter we bring out performances of the path discovery process, considering the number of paths found and the discovery duration beside the amount of overhead generated.

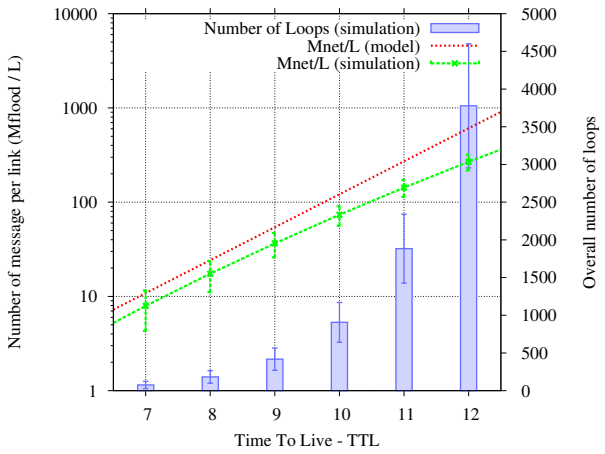


Figure 5: Messages generated per link due to a single flood: model and simulation

Path Discovery Performances

To get an overview of paths discoveries performances (Fig. 4), we pick up 1000 services randomly among the topology: one random node as origin and a distinct random node as destination. Then we launch discoveries from each origin, varying TTL from 7 to 12 (which corresponds here to TTL=D and TTL=D+5). The answer packets arrival to the origin are counted and their arrival time is noticed to respectively get the number of paths discovered and the discoveries durations. Finally, the figure provides the first and the last path discovery duration through the average and standard deviation (respectively through green and red lines), refer to the left axis for durations. The discovered paths are expressed by histograms (read on the right axis). We must also note that the significant standard deviations (both for durations and number of paths) is due to the fact that the random picking of services give a significant variance of the number of hops between enpoints (i.e. their “remoteness") which is itself determining on the amount of paths discovered.

Path Discovery Overhead Cost

To get an overview of the path discovery overhead cost (Fig. 5), we launch this time one path discovery process for each of the 42 nodes

and observe each time the message generation, varying TTL from 7 to 12. The overall generated messages during each flood (i.e. the request messages) are counted. Finally the figure reports the average number of flood messages reduced per link (green line); the previous analytic expectation (see section III.A) is reported for comparison (red line). The simulation confirms that the analytic formula provides an upper bound estimation which is relatively accurate for this network (due to the low standard deviation of nodes degrees, $\sigma_\delta = 1.12$). Yet, it is less accurate when the TTL deviates too much from the network diameter because the amount and size of loops become too significant to be ignored in the analysis. The number of detected loops is reported with the histograms (read on the right axis).

By crossing the Fig. 4 and Fig. 5 results, it is possible to bring out the relevance of TTL settings on the simulated network. For example with TTL=10, path discoveries provide on average hundreds of paths. If we assume for example that such a TTL setting gives sufficient amount of paths for the expected potential of services adaptation and recovery under failure. By looking at the discoveries durations, we see that the last path is obtained around 17ms – whereas the first path is still obtained around 5ms independently from the TTL. Under such discovery durations, a sub 50ms recovery of a broken path can be achieved if faults can be detected between 30ms and 45ms (typically, OAM should be sent every 10ms and a fault considered upon the lake of three packets). On the other hand, in terms of overhead cost, with TTL=10 each flood generates on average almost 70 packets per link.

3.3. *Global Overheads Estimations Over Three Network Profiles*

Services involve floods for their establishment and periodically for their possible adaptations, while active paths are continuously checked by OAM exchanges. We call Optimization Interval (OI) the period of path optimizations, and Service Check Interval (SCI) the period of OAM exchanges. To estimate the OAM overhead, we have to consider the amount of active services (S), and the SCI interval. Yet the OAM message generation also depends on the active paths lengths: if in a network all declared services are using ten hops paths rather than five

hop paths, the OAM exhanges will globally consume twice of the bandwidth. Hence, the exact overhead generated by OAM exchanges among a period of time actually depends on the active paths lengths during this period. We have to make significant assumptions about the active path lengths to get an estimation of the OAM overhead for S services. For a meaningful estimation, we assumed that the services' endpoints "remoteness" distribution (i.e. their shortest paths in number of hops) follows a gaussian distribution probability from 1 to D hops. Then, the effective selected paths can vary over time among the shortest path and the TTL value according to the successive optimizations in the paths selections. By observing that LOCARN active paths lengths rarely exceed the shortest path over two hops through simulations of a realistic transport network topologies and traffic matrix, we are able to give an upper bound estimation of OAM messages per second and per link according to S and SCI.

We summarize in Table 2 the estimations of LOCARN overheads due to non-data packets, gathering estimations of floods overhead based on the formula (2) and on estimations of OAM overhead. Results are reported for the first three network profiles listed in Table 1. For each profile, the LOCARN settings (TTL, OI, SCI) are declined together with the amount of active services across the network. The TTL values are arranged to exceed the network diameter ($TTL = D + 1$, $TTL = D + 2$) providing an important potential for paths optimization. S, OI and SCI are chosen to expose orders of magnitudes ($S = 100/1000$; $OI = 10/30s$; $SCI = 10/100ms$). Finally, the last column reports the sum of overheads estimations per link in ratio beside 1Gb/s that is used here as an indicative basis datarate which allow transpose estimations for higher granularities. The numerical applications confirm that in absolute, the LOCARN overhead due to non-data packets is substancial: on average several mega bit per second and per link for a thousand of services. However, in a Transport Network context (Metro/Core operators parts), such overheads represent few bandwidth consumption. At most, for a thousand of highly resilient ($SCI = 10ms$) and a highly adaptive ($OI = 10s$), the cumulated overheads reach on average 0.85%, 1.98% and 1.50% of 1Gb/s links respectively for profile 1, 2 and 3. Given that nowadays the Transport Networks transmission capacities are commonly around 10, 40 or 100Gb/s in core segments, a such overhead

Network	TTL	S	OI	SCI	Floodings	OAM	Cumulative	Ratio 1Gb/s
Profile 1	12	100	10s	10ms	537Kbps	310Kbps	847Kbps	**0.08%**
				1s		3Kbps	540Kbps	**0.05%**
			30s	10ms	179Kbps	310Kbps	489Kbps	**0.05%**
				1s		3Kbps	182Kbps	**0.02%**
		1000	10s	10ms	5367Kbps	3108Kbps	8475Kbps	**0.85%**
				1s		31Kbps	5398Kbps	**0.54%**
			30s	10ms	1789Kbps	3108Kbps	4897Kbps	**0.49%**
				1s		31Kbps	1820Kbps	**0.18%**
Profile 2	9	100	10s	10ms	212Kbps	1764Kbps	1976Kbps	**0.20%**
				1s		18Kbps	230Kbps	**0.02%**
			30s	10ms	71Kbps	1764Kbps	1835Kbps	**0.18%**
				1s		18Kbps	89Kbps	**0.00%**
		1000	10s	10ms	2116Kbps	17644Kbps	19760Kbps	**1.98%**
				1s		176Kbps	2292Kbps	**0.23%**
			30s	10ms	705Kbps	17644Kbps	18349Kbps	**1.83%**
				1s		176Kbps	881Kbps	**0.09%**
Profile 3	7	100	10s	10ms	1328Kbps	169Kbps	1497Kbps	**0.15%**
				1s		2Kbps	1330Kbps	**0.13%**
			30s	10ms	443Kbps	169Kbps	612Kbps	**0.06%**
				1s		2Kbps	445Kbps	**0.04%**
		1000	10s	10ms	13284Kbps	1688Kbps	14972Kbps	**1.50%**
				1s		3Kbps	540Kbps	**0.05%**
			30s	10ms	4428Kbps	1688Kbps	5916Kbps	**0.59%**
				1s		3Kbps	4431Kbps	**0.44%**

Table 2: Analytic Estimations of non-data packets overheads per link for three network profiles and final ratio to 1Gbps

magnitude remains quite acceptable. Notwithstanding, the bandwidth consumption estimated assumes the packets transfered in both two directions, actually the *directionnal overhead ratio* to 1Gbps corresponds on average to *half of the last column values*.

Finally, the results of this section permit to point out that the actual LOCARN scalability issue (in terms of overheads bandwidth consumption) relies on the amount of active services present simultaneously within a network domain. Indeed, both for routing and OAM, the overall amount of LOCARN's "non-data" packets produced linearly depends on the amount of services. This is the consequence of the LOCARN "service oriented" design in which each service takes its own path decisions and involving consequently its own overheads both for path discovery and for end-to-end fault detection [8].

8. For this reason, the overhead comparison with other protocols and architectures (where this latter only depends on the amount of nodes/links, notwithstanding the usage of statefull nodes) is difficult and may be nonsensical if results interpretation are not achieved carefully.

4. LOCARN's Overhead and Performances Over Failures

As observed in the previous section, floodings due to path discoveries produce numerous packets across the network consuming the network bandwidth accordingly. In section 3 we have thus studied the amount of packets generated, and finally express the overhead cost in terms of mean bandwidth consumption over time. In subsection 3.3, overhead estimations assumed "normal conditions", it is to say that only path optimizations' floodings and OAM exchanges were considered. However, the bandwidth consumption's means values over time (like in Table 2) do not account that floodings occur along very short durations actually depending on the infrastructure propagation delays. Indeed, the floodings do not consume the bandwidth constantly over time, but produce consumption's peaks that must be monitored. Considering the previous section example, if a flooding involves on average 70 packets per link in both two directions (see Fig 5, $TTL = 10$), thus interfaces belonging to the flooding domain will receive around 35 packets during 17ms, which is equivalent to 2060 packets/s.

In addition, the potential problem in LOCARN is that several floodings processes can overlap (i.e. serval flooding processes can be in progress at the same time) which leads to cumulate the peaks of overheads. In normal conditions, given that floodings are launched periodically for each service ("path optimizations"). Given the order of magnitude forseen for optimization intervals in comparison to the floodings duration, the amount of overlapping discovery processes is statistically very low. However, when an infrastructure failure occurs (node or link), if S services are disrupted, S discovery processes are triggered from all the S services source nodes. If the detection time is tuned to very short times (few milliseconds), then the probability of overlapping becomes very high. The overlapping of the S floodings messages generation may cause the filling of packets queues, and even impact the Quality Of Service (QoS) of data transmission, in particular jitter [9]. Yet the impact of floods overlapping is difficult to estimate in a comprehensive way because it may vary a lot according to: (i) the spreading of floods start-

9. as we consider a same priority level both for data and control packets relaying for the sake of simplicty

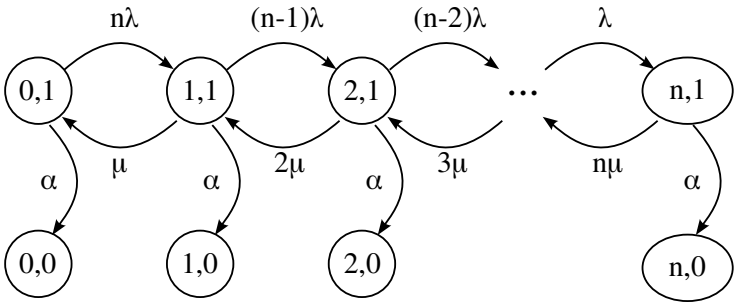


Figure 6: The markov transition diagram – Evolution of services depending on one trail and the risk of its failure

ing over time; (ii) links' propagation delays; (iii) links' available bandwidths during floods propagation.

In this section, we study at first the path recovery that can be expected statistically according to the infrastructure reliability to get a general overview of overlapping floodings according to failures probability. Secondly, we observe by simulations the maximum impact that overlapping recovery processes may involve on the data plane transmission to get an overview of the worst effects of failures.

4.1. *Path Recoveries Statistical Expectation*

To apprehend in a general way the services path recoveries statistical impact, we propose an analytical model to derive the number of services affected by a path failure in LOCARN. Focusing on the path level permits to simplify the modeling process without losing generalities, and hence obtain a trend of the overhead's impacts.

In the envisioned model, we assume that: (i) the inter-arrival time of each service follows an exponential distribution with rate λ; (ii) the service duration follows an exponential distribution with rate μ. In the same way, the time duration before a path failure occurs is exponentially distributed with rate α. We consider that the maximum number of services supported by a LOCARN path is n. The above assumptions conduct us to model the system using a Markov Chain $X = \{X_t, t \geq 0\}$ on the state space S defined by $S = \{(i,k)|i = 0,\ldots,n,\ and\ k = 0,1\}$,

for every $n \geq 1$. $X = (i, j)$ means that, at time t, there are i active services in the path and the latter is in state k. $k = 1$ indicates that the path is ok, while $k = 0$ means that the path is in failure state. Fig. 6 shows the transitions graph of the system. We remark that all states where $k = 0$ (failure) are absorbing states. Such design permits to know the number of active services when the system enters an absorbing state. Here, we focus only on capturing the system state when it fails. The case of reestablishing the path connection is out of scope of this work.

The different transitions of this chain are as follows:

– If a service arrives while already i ($0 \leq i \leq n-1$) services are active and the path is ok, then there is a transition from state $(i, 1)$ to $(i+1, 1)$ with rate $(n-i)\lambda$.

– If a service leaves while already i ($0 \leq i \leq n-1$) services are active and the path is ok, then there is a transition from state $(i, 1)$ to $(i-1, 1)$ with rate $(i)\mu$.

– If a path failure occurs while already i ($0 \leq i \leq n-1$) services are active then there is a transition from state $(i, 1)$ to $(i, 0)$.

We denote by Q_B the transition matrix between non-absorbing states. It is worth noting that this matrix does not represent the infinitesimal generator of this chain. Let σ_B be the initial probability distribution vector of the chain states, and $Q_{B,k}$ the vector containing the transitions rate from the non-absorbing states to the absorbing state i. Q and $Q_{B,1}$ could be obtained as follows:

$$Q_T = \begin{bmatrix} -(n\lambda+\alpha) & n\lambda & 0 & 0 & \dots & 0 \\ \mu & -((n-1)\lambda+\alpha+\mu) & (n-1)\lambda & 0 & \dots & 0 \\ 0 & 2\mu & -((n-2)\lambda+\alpha+2\mu) & (n-2)\lambda & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \dots \\ \vdots & \vdots & \vdots & \vdots & n\mu & -(\alpha+n\mu) \end{bmatrix}$$

$$Q_{B,1} = \begin{bmatrix} \alpha \\ 0 \\ 0 \\ \vdots \\ \vdots \end{bmatrix}$$

The probability π_k to be in the absorbing state k (i.e. the state (k-1,0) of our notation)) is obtained as follows:

$$\pi_{k-1,0} = -\sigma_B(Q_B)^{-1}Q_{B,k}$$

We note the vector v as $v = -\sigma_B(Q_B)^{-1}$. Knowing that the initial state from which the system begins is state (0,1) and by deduction we can obtain the linear system $vQ = -\sigma_B$ that can be written as follows:

$$\begin{cases} -(n\lambda+\alpha)v_0 + \mu v_1 = -1 \\ (n-i+2)\lambda v_{i-2} - ((n-i+1)\lambda + (i-1)\mu + \alpha)v_{i-1} + i\mu v_i = 0 \quad for\ i = 2\ to\ n \\ n\mu v_{n-1} - (n\mu+\alpha)v_n = 0 \end{cases}$$

Knowing that $\pi_k = \frac{v_k}{\alpha}$ (since $\pi_k = vQ_{B,k}$) and replacing its value in the precedent linear system we obtain:

$$\begin{cases} -(n\lambda+\alpha)\pi_0 + \mu\pi_1 = -\alpha \\ (n-i+2)\lambda\pi_{i-2} - ((n-i+1)\lambda + (i-1)\mu + \alpha)\pi_{i-1} + i\mu\pi_i = 0 \quad for\ i = 2\ to\ n \\ n\mu\pi_{n-1} - (n\mu+\alpha)\pi_n = 0 \end{cases}$$

We replace the last equation by the normalizing condition $\pi\mathbb{1} = 1$ we obtain:

$$\begin{cases} -(n\lambda+\alpha)\pi_0 + \mu\pi_1 = -\alpha \\ (n-i+2)\lambda\pi_{i-2} - ((n-i+1)\lambda + (i-1)\mu + \alpha)\pi_{i-1} + i\mu\pi_i = 0 \quad for\ i = 2\ to\ n \\ \pi_1 + \pi_2 + \pi_3 + \cdots + \pi_n = 0 \end{cases}$$

Accordingly we have the following recurrence relation:

$$\begin{cases} \pi_1 = \frac{(n\lambda+\alpha)}{\mu}\pi_0 - \frac{\alpha}{\mu} \\ \pi_i = \frac{((n-i+1)\lambda+(i-1)\mu+\alpha)}{i\mu}\pi_{i-1} + \frac{(n-i+2)\lambda}{i\mu}\pi_{i-2} \quad for\ i = 2\ to\ n \end{cases}$$

To solve this recurrence, we start with any positive value of π_0, and then we calculate all values of π_i (for $i = 1, \ldots, n$). Then, we obtain the real values of π_i by dividing each value by the sum of π_i. Having calculated the different probabilities (π_i), we are able to compute the expected number of active services ($E[S]$) on the path when a failure occurs:

$$E[S] = \sum_{i=0}^{n+1}(i-1)\pi_{(i-1,0)} \tag{3}$$

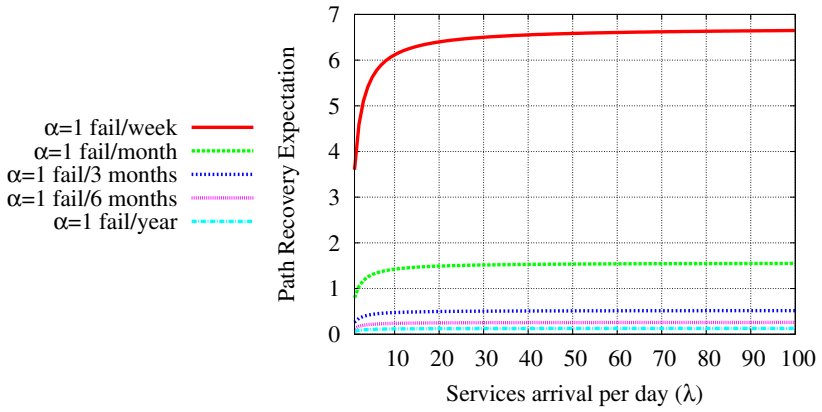


Figure 7: Expectation of services disconnections per day due to the path failure ($n = 50$)

In fact, $E[S]$ gives the expected number of active services in the path when a failure occurs. Knowing that each service generates M messages in order to find another path, the expected number of messages generated in this case $E[M]$ is:

$$E[M] = ME[S] \tag{4}$$

Usually, we are interested on the number of messages generated during a period of time. We know that the mean time before absorption (i.e. failure) of the modeled system is $\frac{1}{\alpha}$. Therefore, the expected number of messages generated during a period (say day, if α represents a failure rate by day) is equal to:

$$E[MPeriod] = \alpha E[M] \tag{5}$$

Results and Observations

At this point, we are able to get the number of recovery processes that can be expected when λ, μ and α are known with a maximum of n fixed services by using (3). All the results thereafter assume that services mean duration before leaving is on average $\frac{1}{\mu} = one\ day$, knowing that what is finally impacting is the ratio $\frac{\lambda}{\mu}$. In Fig. 7, by using

	$\lambda =$ \ $\alpha =$	$\frac{1}{year}$	$\frac{1}{6months}$	$\frac{1}{3months}$	$\frac{1}{month}$	$\frac{1}{week}$
$n = 20$	$\frac{10}{day}$	0.042	0.086	0.172	0.516	2.215
	$\frac{30}{day}$	0.045	0.091	0.183	0.548	2.351
	$\frac{100}{day}$	0.046	0.093	0.187	0.561	2.395
$n = 50$	$\frac{10}{day}$	0.117	0.237	0.475	1.425	6.116
	$\frac{30}{day}$	0.125	0.253	0.505	1.516	6.500
	$\frac{100}{day}$	0.127	0.259	0.517	1.551	6.647
$n = 100$	$\frac{10}{day}$	0.242	0.490	0.980	2.941	12.618
	$\frac{30}{day}$	0.257	0.522	1.043	3.129	13.414
	$\frac{100}{day}$	0.263	0.534	1.067	3.201	13.721

Table 3: Amount of paths recovery launches expected (per day)

realistic transition rates magnitudes, we provide the expectation of services recovery processes launched per day ($E[S/day] = \alpha E[S]$) for $n = 50$ in Fig. 7. Once the maximum amount of maximum services n is reached by the markov process, the services disconnections expected almost only depends on n and α. It can be observed that for the choosen α and n values, the ratio $\frac{\lambda}{\mu}$ has a noticeable impact on expectation until 10. Those permit to see that, upon realistic settings, the significant impacting factors are α and n. To get a larger overview, some result are gathered in Table 3 where the same observations can be made.

To exhibit more convenient values, it is possible to apply the analytic formula (2) from section 3.1 to determine M, which allows us to find finally the messages expected per link and per day by considering realistic network dimensions and LOCARN settings (i.e. δ_{net}, D and floodings' TTL). Doing so, we provide in Table 4 some estimations of path recoveries overhead that can be expected, based on both realistic transition rates and network dimensions. Network dimensions assumed are the same than in the section 3.2 simulations, we assume the France optical national backbone (i.e. N=42 nodes, L=68 links, diameter D=7 hops, the mean density $\delta_{net} = 3.23$ and the nodes degree standard deviation is $\sigma_\delta = 1.12$). Hence, to get a consistant view for transport networks estimation, we can specifically look at results with $\alpha = \frac{1}{year}$ or $\alpha = \frac{1}{6months}$. Here we see that in terms of packet expectation per day, the path recovery overheads are unsignificant since the maximum of services n is acceptable.

	$\lambda =$ \ $\alpha =$	$\frac{1}{year}$	$\frac{1}{6months}$	$\frac{1}{3months}$	$\frac{1}{month}$	$\frac{1}{week}$
$n = 20$	$\frac{10}{day}$	1.722	3.492	6.984	20.964	90.075
	$\frac{30}{day}$	1.833	3.717	7.433	22.300	95.580
	$\frac{100}{day}$	1.875	3.802	7.602	22.794	97.390
$n = 50$	$\frac{10}{day}$	4.760	9.653	19.308	57.941	248.694
	$\frac{30}{day}$	5.067	10.275	20.551	61.656	264.322
	$\frac{100}{day}$	5.184	10.512	21.025	63.074	270.304
$n = 100$	$\frac{10}{day}$	9.824	19.922	39.847	119.570	513.065
	$\frac{30}{day}$	10.458	21.206	42.413	127.244	545.445
	$\frac{100}{day}$	10.699	21.696	43.392	130.176	557.917

Table 4: Amount of packets expected (per link and per day)

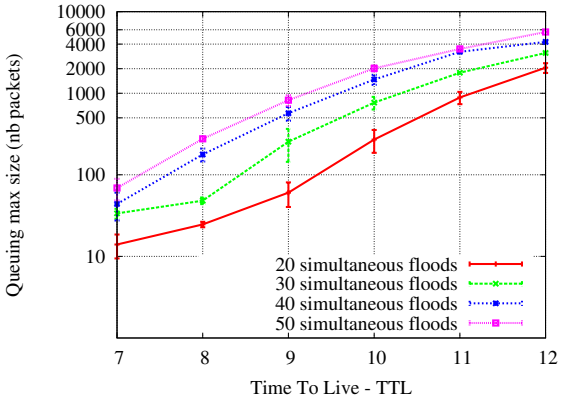
4.2. *Path Recovery Maximum Impact*

To bring out some meaningful estimations about the concrete impact of overlapping floodings, we consider a very worst case scenario and distinguish two impact factors: the amount of overlapping discovery proccesses, and the floods propagation scope (TTL). In order to observe the impact of floodings on network data transmissions, we proceed as follows: (i) first we assume the same network than presented in section 3.2 (N=42 nodes, L=68 links, diameter D=7 hops, the mean density $\delta_{net} = 3.23$ and the nodes degree standard deviation is $\sigma_\delta = 1.12$) ; (ii) then all links capacities are set to 1Gb/s that we use as an indicator like in Table 2 ; (iii) a constant data traffic is set over a path identified as "especially at risk" [10] in the network up to two-third of capacities ; (iv) then we launch *simultaneously* [11] S floodings processes (with $S = 20, 30, 40, 50$). Floodings origins are picked out randomly, and simulation are lauched 20 times for each S. We also make TTL vary from $TTL = D = 7$ to $TTL = D + 5 = 12$.

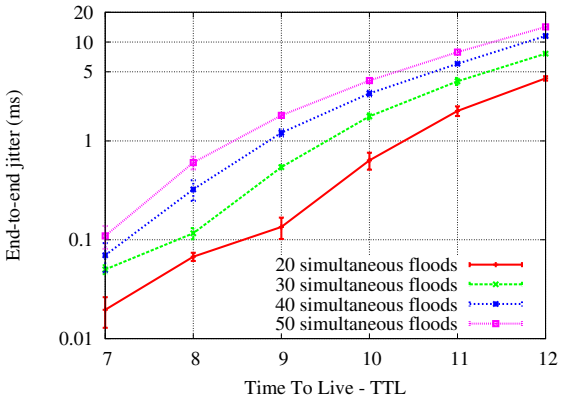
Then the impact is measured by observing the studied path: the maximum queue length that have been reached along flooding sessions can be noticed on one hand and the end-to-end jitter (by noticing the worst

10. such a path have been noticed sensitive to flooding perturbations over several simulations: it is a long path passing through the dense network zone

11. the fact than N floods are launched strictly at the same time actually corresponds to a very worst case that statistically should never happen. This allows us to bring out an upper bound of impacts on the network.



(a) Queues maximum length



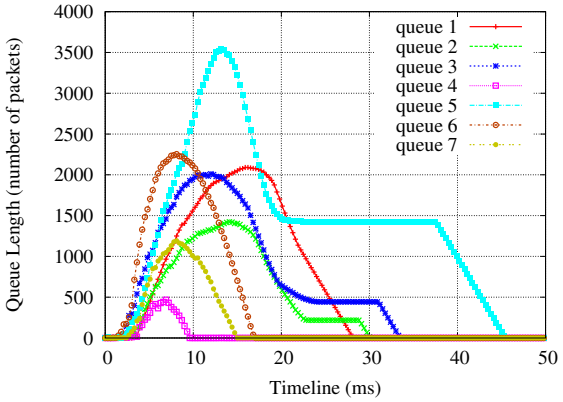
(b) Worst packet jitter

Figure 8: Impact of simultaneous floods on the studied path (1Gbps links)

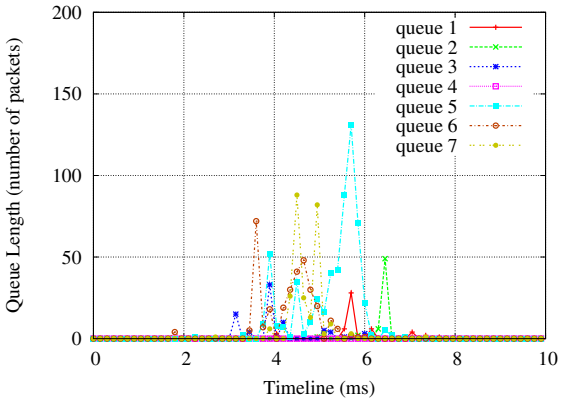
packet delay observed) on the other hand. To bring out impacts independently from the starting points, we report results' averages and standard deviations over 20 simulations.

Results and Observations

With Fig. 8, we exhibit the impact of N overlapping floods in terms of queues load and jitter whereas in Fig. 9 we observe the queues length evolution over time along the previous sensible path in the case all nodes (42) would launch a flood at the same time, in case of recovery for



(a) With 1Gbps links



(b) With 10Gbps links

Figure 9: Evolution of queing loads along the studied path (42 simultaneous recoveries, TTL=12)

example. The figures depict the evolution of queues lengths over time along the studied path. Respectively Fig.9a and Fig.9b permit to show how much the impact is reduced by passing from 1Gb/s to 10Gb/s links.

Finally, the results of this section permit to point out that, for transport networks having highly reliable infrastructures based on Fiber Optics transmission, the really impacting factor of scalability remaining is the amount of services to repare when a failure occurs. This correspond to the n parameter in the statistical modeling and to the number

of simultaneous floodings launched at the same time in simulations. Moreover, the network ability absorption to cope with bursts in very extreme scenarios on the simulated examples both depends on its occupancy rate and on its transmission capacities. Let us note however that beyond 10Gb/s links, tens of *exactly simultaneous* recoveries are almost imperceptible for the data plane transmissions.

5. Conclusion

In this article, we have presented and studied LOCARN: an alternative packet architecture providing point-to-point communications through a very simple design and concepts. After the technical overview and analysis in the section II, the article focuses on scalability and performances of the architecture, considering in particular the overheads production involved by routing and OAM processes which constitute the critical technical aspects. Overheads and performances studies are treated seperately between “normal" and “over failure" contexts through sections III and IV. Finally, both the two sections lead to the conclusion that the decisive factor of the LOCARN scalability (i.e. impacting its performances) is the amount of services present within the domain. This conclusion logically results as the counterpart of its very singular “service oriented" design. Based on this observation, we are currently proposing some architetural improvements that will be published soon. For now, the results already permit to say that LOCARN is widely acceptable for mesh networks with high datarates ($\geq 1Gbps$) until thousands of communications and with very adaptive and resilient settings. Therefore we think that LOCARN constitutes a mean for operators to answer the increasing needs for flexibility and dynamicity in their core networks without complexe configuration, protection settings, and even plannification steps. Doing so it allows them to simplify both the operationnal investment and the network devices complexity, it is to say to reduce drastically both OPEX and CAPEX financial costs.

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